Summer Examinations 2017-18

Exam Code(s)	3BME1, 4BME1, 1EM1, 1OA1, 1OA9, 4BCT1, 4BMS2, 3BS9, 4BS2
Exam	Third Year & Fourth Year & HDip
Module	CRYPTOGRAPHY
Module Codes	CS402 & MA492 & MA545
External Examiner(s)	Prof T. Brady
Internal Examiner(s)	Prof G. Ellis*
Instructions	Attempt **all** questions. There are ten questions, each carrying 10 marks.
Duration	2 hours
No. of Pages	3 pages (including this cover page)
Discipline	Mathematics

Requirements:

Release in Exam Venue	Yes ☑	No ☐
Release to Library	Yes ☑	No ☐
Non-programmable calculator	Yes ☑	No ☐
Mathematical Tables	Yes ☐	No ☑

1. The ciphertext

$$CQRBRBJFNJTLRYQNA$$

was produced with a Caeser cipher over the alphabet $A = 0, B = 1, ..., Z = 25$. Frequency analysis has been used to determine that E gets enciphered as N. Determine the **first word** of the corresponding plaintext.

2. The ciphertext

$$ESDCWNMH$$

was produced by applying a Hill cipher to 2-letter message units over the alphabet $A = 0, B = 1, ..., Z = 25$, with enciphering finction

$$f_E\colon \begin{pmatrix} x \\ y \end{pmatrix} \mapsto A \begin{pmatrix} x \\ y \end{pmatrix} \qquad \text{where } A = \begin{pmatrix} 2 & 3 \\ 7 & 5 \end{pmatrix}.$$

Calculate A^{-1} mod 26 and hence determine the **first two letters** of plaintext.

3. Describe:

(a) *Kerckhoff's principle.*

(b) the difference between *symmetric cryptography* and *public key cryptography.*

(c) one basic property of the Enigma cipher which makes it significantly more secure than the Vigenère cipher against a ciphertext only attack.

4. Determine, with explanation, the size of the enciphering key space for an affine cipher $f_E\colon (\mathbb{Z}_p)^d \to (\mathbb{Z}_p)^d, v \mapsto AV + B$ over an alphabet of p letters with p a prime.

5. A 4-bit linear feedback shift register with connection polynomial $C(X) = 1 + X + X^3$ is used to produce a pseudo-random sequence of binary digits, starting $s_0 = 1$, $s_1 = 0$, $s_2 = 1$, $s_3 = 1$. The sequence is eventually periodic. List the next few terms of the sequence in order to determine its period.

6. Answer **either** part (a) **or** part (b).

(a) Explain the following terms: (i) *two factor authentication*; (ii) *forward security*; (iii) *Carmichael number.*

(b) Let $\mathbb{P}$, $\mathbb{C}$, $\mathbb{K}$ denote respectively the plaintext space, ciphertext space and key space of a given cryptosystrem. Assuming that each of these spaces is finite, that the cryptosystem is perfectly secure and that every $c \in \mathbb{C}$ has non-zero probability of occuring, prove the inequalities

$$|\mathbb{K}| \geq |\mathbb{C}| \geq |\mathbb{P}|.$$

7. Answer **either** part (a) **or** part (b).

(a) The enciphering key for a binary stream cipher is produced using a 3-bit linear feedback shift register. It is known that the cipher converts the plaintext string

$$0010001101010111$$

into the ciphertext string

$$1000010000011001 \; .$$

Determine the first sixteen binary digits in the enciphering key. Then determine the connection polynomial of the linear feedback shift register.

(b) Carefully describe the A5/1 stream cipher which is used to encrypt the on-air traffic in the GSM mobile phone networks in Europe and the US.

8. Answer **either** part (a) **or** part (b).

(a) State Euler's Totient Theorem and use it to prove that

$$(a^e)^d \equiv a \bmod N$$

for $N = pq$ with p and q distinct primes, e an integer with $gcd(\ e,\ (p-1)(q-1)\ ) = 1$, a any integer such that $gcd(a, N) = 1$ and $d \equiv e^{-1} \bmod (p-1)(q-1)$.

(b) Describe Fermat's primality test (which is passed by every prime). Suppose that m is a random composite number which is not a Carmichael number. Prove that m passes the test k times with probability $\leq (1/2)^k$.

9. Answer **either** part (a) **or** part (b).

(a) Alice and Bob choose the abelian group $\mathbb{Z}_{71}^*$ and generator $g = 7$ to perform Diffie-Hellman key exchange. Alice secretly chooses $a = 6$ and Bob secretly chooses $b = 13$. What is the numerical value of their shared key?

(b) Let E denote the set of points on the elliptic curve over $\mathbb{Q}$ defined by $y^2 = x^3 + ax + b$. Describe (without giving precise formulae) the operations of addition and subtraction which give E the structure of an abelian group.

10. Answer **either** part (a) **or** part (b).

(a) Describe Pollard's rho method by using it to factorize $n = 91$ with $f(x) = x^2 - 1$ and $x_0 = 2$. (Compare x_k with x_j for which $j = 2^h - 1$ where $2^h \leq k < 2^{h+1}$).

(b) Describe one perfectly secure (though possibly impractical) cryptosystem. By carefully stating and using a theorem of Shannon, or otherwise, explain why the cryptosystem is perfectly secure.